

Revisiting Engagement in Humanitarian Mapping: An Updated Analysis of Contributor Retention in OpenStreetMap

Zarif Mahmud
University of Toronto
Toronto, Canada
zarif.mahmud@mail.utoronto.ca

Dipto Sarkar
Carleton University
Ottawa, Canada
dipto.sarkar@carleton.ca

Aarjav Chauhan
University of Toronto
Toronto, Canada
aarjav.chauhan@mail.utoronto.ca

Robert Soden
University of Toronto
Toronto, Canada
soden@cs.toronto.edu

ABSTRACT

This study revisits findings of the 2016 paper by Dittus et al. that considered trends in contributor retention in humanitarian mapping projects organized using the OpenStreetMap (OSM) platform. In addition to revisiting many of the same metrics used for the 2017-2020 time period, our research takes on a broader scope by evaluating a wider cohort of recruits, as opposed to specific projects evaluated by the original paper. As a result, our findings offer an updated and more complete picture of contributor retention in humanitarian mapping projects. We also offer several insights and future research opportunities for long-term contributor retention in online peer production projects.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing**; *Collaborative content creation*; **Empirical studies in collaborative and social computing**; • **Information systems** → *Crowdsourcing*.

KEYWORDS

OSM, retention rates, open-data

ACM Reference Format:

Zarif Mahmud, Aarjav Chauhan, Dipto Sarkar, and Robert Soden. 2022. Revisiting Engagement in Humanitarian Mapping: An Updated Analysis of Contributor Retention in OpenStreetMap. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3491101.3519728>

1 INTRODUCTION

This paper examines contributor retention on the OpenStreetMap (OSM) platform. OSM is a project that focuses on "collaborative mapping", creating an open world map that can be edited and

contributed to by people around the world. Its goals and scope are similar to Wikipedia, in that it seeks to create a decentralized method to update a body of knowledge, that can then be used freely by anyone else in the world. It was founded in 2004 in the UK, but has grown into a massive worldwide project with over 1 million unique user accounts who have collectively contributed billions of changes and updates to the global map. These changes can range from marking existing roads and houses, to updating developing areas as new buildings or structures are constructed, to even updates not found in most maps, such as speed limits. As with other online collaborative projects, like Wikipedia or open-source software projects, contributor retention is a crucial problem. Maintaining a group of long-term users for these sites is crucial for their overall success and health for various reasons, such as experienced contributors being able to disseminate knowledge to newer users, as well as shaping the overall culture and direction of these sites.

This work revisits research by Dittus et al. from 2016, where the authors analyzed recruitment during three different humanitarian mapping and recruitment drives, and examined various factors that may have led to recruits staying on or leaving after the drive [5]. We extend this work by evaluating if those findings still hold up for the present day, as OSM has grown and expanded in the intervening years, as well as taking a broader look at these individual projects beyond particular drives. Following the original paper, we focus our research on humanitarian elements of OSM. These projects, which seek to harness volunteer contributions to support disaster response and international development agencies through provision of map data, have been a major source of growth in OSM in recent years [9]. Two projects that we will focus on in particular are Missing Maps and YouthMappers.

Missing Maps is a non-profit organization that works within OSM. It was founded in November 2014 as a joint effort between various humanitarian NGOs (the American Red Cross, British Red Cross, Humanitarian OpenStreetMap Team, and Doctors Without Borders). Its goal is to map underserved areas where people may face natural disasters, disease outbreaks, or other crises so that communities and outside aid teams can better render support. To date, there have been over 100,000 unique contributors from around the world who contributed to various projects to map underserved parts of the world, and have helped map large swathes of many previously under-mapped countries, such as in the Democratic

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-9156-6/22/04...\$15.00

<https://doi.org/10.1145/3491101.3519728>

Republic of the Congo and South Sudan [5]. Missing Maps regularly holds mapping drives and "mapathons" in order to crowd-source the necessary labour needed to map large swathes of area. We also analyzed the YouthMappers project. YouthMappers is a network of local university chapters around the world, and many of its goals are similar to that of Missing Maps (in fact, it is currently a partner to Missing Maps). It also hosts mapping drives in order to map various underserved parts of the world, but it differs from Missing Maps in that its goals are more broadly to empower younger students with the tools to map their communities, and build grassroots movements where they are. [13]

In this paper we seek to evaluate which aspects of contributor recruitment in humanitarian groups that are working with the OSM platform may be effective for retaining users. In our analysis, we found that the structures and incentives driving these various organizations may be a big source of differences in retention rates. We find evidence of a potential tradeoff between being able to mobilize a lot of new volunteers quickly and being able to retain long-term contributors, as can be seen when comparing the humanitarian groups' efforts to the overall community of contributors to OSM.

2 RELATED WORKS

To understand contributor retention in humanitarian projects in OSM, Dittus et al. [5] evaluated past OSM projects to figure out which factors may lead to more engagement and retention. The projects analyzed were "mapathons" organized for Typhoon Yolanda, the Ebola outbreak in Guinea, as well as the Missing Maps project. Mapathons are short, time-bound events where participants receive training in how to contribute to the OpenStreetMap platform, usually through tracing roads and buildings found in satellite imagery, and are then asked to work on areas of interest where map coverage is limited. The paper used various metrics, such as analyzing short and long-term retention and task analysis, to assess predictors of a contributor staying on for the long-term. Short-term retention was measured using the amount of recruits who returned the second day after their initial join date, while long-term retention was measured based on the amount of recruits who returned in Month 2 and 3 after initial join date. The paper found that Missing Maps had the lowest amount of short-term retention, while the two other projects, Typhoon Yolanda and Ebola Response, performed better. However, in contrast, Missing Maps had the most stable long-term retention rates, with only a small percentage of recruited mappers leaving after two months. Other metrics considered were labour hours and contribution rates, in which Missing Maps recruits tended to work the longest hours, but had a lower rate of contribution than the other two projects.

In the years following the paper by Dittus et al., the number of recruits to humanitarian OSM projects such as YouthMappers and Missing Maps have steadily increased, such that the amount of new recruits is many times the amount it was in 2016. OSM overall has also continued to grow, with new contributors and users of the map data continuing to join. Although the motivation for this paper was mainly to update the findings of Dittus et al., a sort of "Where are they now?" for the metrics used there, this work also builds off of a rich literature of analyzing metrics of contributor engagement. In the space of analyzing engagement for volunteers

of open-source projects, Wikipedia is a widely studied site. One of the metrics used in the Dittus et al. paper and our own is the idea of "labour hours" which was adapted from Geiger's metric of analyzing Wikipedia users' edit sessions [6]. Other recent studies focus on the demographics of participants, and how barriers to participation may exacerbate "digital inequalities" [11].

Qualitative studies have also contributed to our understanding of contributor retention in OSM. Budhathoki [2] surveys "serious" and "casual" contributors to OSM to see what the different motivations for contributing may be; serious mappers tended to join for reasons like community and career advancement, while casual mappers tended to join for general interest around open-source mapping. Related, Cox et al. [4] found that Wikipedia contributors motivated by the values of an online volunteer project, as opposed to their own social or career advancement, tended to have the highest level of retention and engagement. Recent work has also analyzed the "life cycles" of a typical OSM contributor in Bégin et al. [3], which identifies trends in the volume and frequency of contributions over time.

3 METHODS

3.1 Data

Data on contributor participation in OSM was collected and analyzed using the Amazon Athena platform. Athena is a cloud-based database query service that allows OSM data to be analyzed and accessed using SQL, without needing to store the data locally. OSM contains editing data going back to 2007, stored in a PostgreSQL database. The size and format of that database makes it prohibitively difficult to analyze on its own [1], so using Athena makes the massive amounts of data more amenable to analysis. The primary type of data we evaluated was a "changeset". Changesets are groupings of individual edits to the map, created whenever a user commits their changes to the map. A changeset may contain up to 10,000 edits, though the majority contain only a handful. Additionally, a changeset contains valuable metadata such as a user's unique ID and when the changeset was made, as well as tags and comments that help identify if the contribution was added as part of a larger project. This allowed us to perform queries to identify when a user had joined, how often they contributed, what project they were likely to have been recruited from, the frequency of their contributions, and more. These are important, as they give a window into the contribution and editing habits of users, and insight into what may encourage users to stay and contribute for the long-term.

3.2 Study Period and Cohorts Analyzed

The main groups studied in this paper are recruits of Missing Maps, YouthMappers, and the general cohort of all OSM members who joined during the 2017-2020 time period. We thus extended the analysis from the Dittus et al. paper to this period, as well as expanded the range of cohorts studied. 2021 wasn't included as it was still too recent to accurately analyze long-term retention rates. The original paper focused on specific mapping drives to see recruitment, while this paper places more focus on overall trends within the three groups over the entire time period. We also included the general cohort to have a baseline against which to compare the humanitarian organizations.

Table 1: Short and long-term retention from 2017-2020 (number of users and percentage of users remaining)

	Beginning	Day 2	Month 2	Month 3
MissingMaps	86033 (100%)	4277 (5.0%)	4485 (5.2%)	3185 (3.7%)
YouthMappers	8972 (100%)	646 (7.2%)	527 (8.3%)	451 (5.0%)
All	811200 (100%)	63925 (7.9%)	68422 (8.4%)	49153 (6.1%)

Table 2: Short and long-term retention rates from 2013 - 2016 (number of users and percentage of users remaining)

	Beginning	Day 2	Month 2	Month 3
MissingMaps	16609 (100%)	885 (5.3%)	830 (5.0%)	716 (4.6%)
YouthMappers	1662 (100%)	112 (6.7%)	149 (9.0%)	66 (4.0%)
All	805093 (100%)	79878 (10.0%)	79003 (9.8%)	58225 (7.2%)

This paper recreates the parts of the analysis of the original paper that are applicable to these new cohorts, as well as adding additional metrics, such as analyzing whether initial frequency of contribution leads to higher retention prospects. We evaluate how the findings of the original paper hold up, with the amount of new users to OSM greatly increasing and also the relationship between users and the sites they contribute to rapidly changing, in order to contribute to a broader understanding of user retention in online communities.

Due to the different types of cohorts we are considering compared to the original paper, not all of the original metrics make sense to replicate here. For example, quantifying prior OSM experience makes sense when considering recruits on a per short-term drive basis (e.g. a mapathon or series of mapathons focused on a particular crisis). But as we're considering overall recruitment to various organizations, focusing on recruits who are not just new to the project, but to OSM in general, this metric was not relevant.

3.3 Metrics

To identify a "recruit" to Missing Maps or YouthMappers, we looked at the hashtags submitted with each changeset. Organizations such as MissingMaps and YouthMappers automatically add hashtags to user submissions on projects ran by them. If a hashtag related to Missing Maps or YouthMappers is within a user's first two contributions to OSM, they are considered a recruit of that project. Allowing for the hashtag to be in a user's second changeset allows for an initial "test" contribution, as new recruits to OSM projects often make a small, non-project related change as part of introductory tutorial activities (such as mapping their own house or workplace), before beginning the mapping requested by the recruiting organization.

Analysis of short vs. long-term retention was modelled on the metrics used in the original Dittus et al. paper, an approach known as "survival analysis". For all recruits within the three groups, data was gathered on all of the changes they had contributed to OSM within the first 180 days of joining. First, we considered the first 48 hours to look at short-term retention. In order to analyze long-term retention, we looked at contributions made in the second and third months after joining. Only the first 90 days would be used for the direct analysis of retention, while the 90 days afterwards would be included to help model overall survival. For the survival analysis, a contributor is considered "dead", if they haven't contributed in the

90 days after the first 90 days already considered (i.e. Day 91-Day 180 after initial recruitment). Using the full 180 day data, survival curves were plotted to show the likelihood of any user staying after a given amount of days. Because the cohorts selected in this analysis somewhat differ from the cohorts analyzed in the original paper, these metrics were also run on the selected cohorts of this study from 2016 and prior, in order to see where they may differ from the original results of Dittus et al., and to make better comparisons with our current numbers.

Another metric considered in the initial paper was whether the amount of "labour hours" a person performed within the first 48 hours had any impact on their overall retention rate. Labour hours are measured by "a sequence of edits made by an editor where the difference between the time at which any two sequential edits are saved is less than one hour" [6]. Labour was also measured by the average amount of edits contributed per hour within the first 48 hours. Lastly, a logistic regression model was used on the YouthMappers cohort data to analyze whether there was a relationship between the number of edits a person made on their first day using OSM, with their probability of "surviving" until the end of the time period.

4 RESULTS

The first metric analyzed was the short and long-term retention rates of the different cohorts. Both time periods (2017-2020 and 2016 and prior) were considered for this analysis. After collecting first-time contributors, the retention rates for people who returned within 48 hours, the second month, and the third month, were calculated (Table 1). These results show that the selected cohorts have lower retention rates (both short-term and long-term) than the overall OSM user base, suggesting that projects that add to the OSM map using crowd-sourcing volunteers have lower retention rates. Table 2 presents the same analysis, but performed within the time frame assessed by the original paper. These results largely mirrored the results from the 2017-2020 period, but with higher overall retention across most time frames (48 hours, 2 months, and 3 months) for each cohort.

Figure 1 shows the survival rates of each cohort 90 days after the initial 90 days from when a person started mapping on OSM. If someone had not contributed a change in the latter 90 days (from Days 91 - 180), they were considered to be "dead" or inactive for

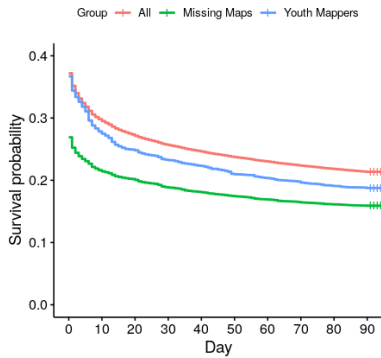


Figure 1: Survival Rates of Cohorts

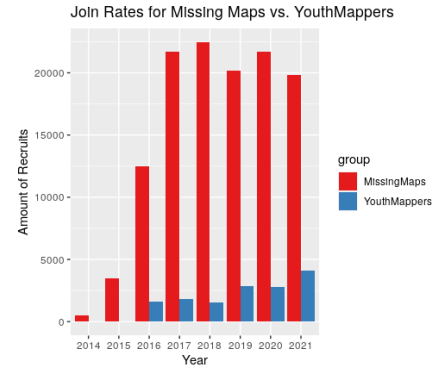


Figure 2: Overall Join Rates for Missing Maps and YouthMappers

Table 3: Average Labour Hours and Contribution Rates for Cohorts (first 48 hours)

	Average Labour Hours	Average Contribution Rates
MissingMaps	0.90 hours	4.1
YouthMappers	0.74 hours	3.0
All	0.55 hours	2.2

the purposes of the analysis. The results show that rates of decay are not too different between the cohorts, but MissingMaps' curve starts from a lower overall survival probability, reflecting their lower retention in comparison to the other two cohorts.

Figure 2 shows the overall recruitment by Missing Maps and YouthMappers since their inception (users who've had a hashtag from one of these groups within their first two contributions). An important conclusion from this graph is that the amount of new recruits to Missing Maps has greatly increased since 2016, such that the amount of new recruits from 2017-2020 dwarfs the amount from 2016 and prior, further highlighting the need for an updated analysis on contributors within OSM.

Next we calculated the average labour hours for each cohort in the first 48 hours, as well as the average contribution rate, which was measured by the average number of edits a user made per hour in the first 48 hours after joining. These calculations were conducted for the 2017-2020 time period. Table 3 presents these results, where we see that the average labour hours and contribution rates for Missing Maps significantly exceeds the other two cohorts within the first 48 hours. This finding differs from the results from original Dittus et al. study, where amongst the cohorts studied, Missing Maps had the lowest contribution rate. Though Missing Maps also had the most labour hours amongst the cohorts in the original study, our data suggests increased initial contributions in comparison to Youth Mappers and the general cohort. Overall, the average labour hours in the first 48 hours for the cohorts studied in this analysis was less than in original work by Dittus et al., where the average labour hours were all between 1-1.3 hours, while in this analysis they fall between 0.5-1 hour.

Lastly, we analyzed whether the rate of initial contribution had any effect on the overall likelihood of still being active during the survival period. We used logistic regression to model this on the

YouthMappers cohort data, with the binary variable being a flipped version of the survival metric used before (i.e. a user is assigned a "1" if they are active in Days 91-180 after initial recruitment, 0 otherwise). This models the overall probability of a user being active within this time period, dependent on the number of contributions made. Shown in Figure 3, we see a slight relationship between the amount a person contributes on their first day (on the x-axis), and their overall survival chances after 180 days. This raises the possibility that encouraging users to contribute more in their first few days may encourage them to stay engaged with the project for a longer period of time.

Contributor Retention vs. Number of Contributions

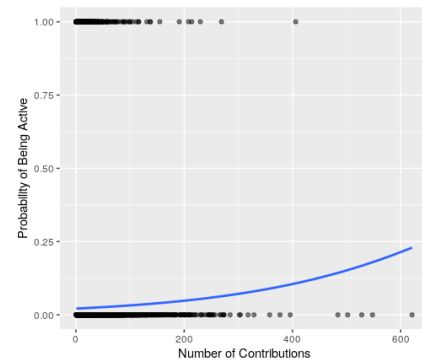


Figure 3: In this figure, the black dots indicate if a user "survived" within the Day 91-180 survival period, i.e. they contributed at least one edit. This survival is mapped against the amount of contributions a user made within the first 24 hours after recruitment.

5 DISCUSSION

Due to our different approach, our findings somewhat diverge from the conclusions presented by Dittus et al. In contrast to the original paper's findings that Missing Maps had low short-term retention but the best long-term retention rates, data from 2017-2020 finds that Missing Maps has both lower short and long-term retention rates when compared to both YouthMappers and the general OSM population. This does not just apply to the recent data; the analysis on the 2016 and prior data also yielded similar results, as seen in Table 2. These differences can be explained by the different types of cohorts analyzed. Dittus et al. considered specific focused drives, such as the response to Typhoon Yolanda, where users may be more likely to return on the second day (as mapathons are often multi-day events, especially around time-critical events such as a typhoon). In addition, when looking at the broader "general" OSM cohort, we see that humanitarian projects have the lowest retention overall.

One reason for this might simply be the nature of the incentives of people who join OSM through Missing Maps versus other groups. Missing Maps normally recruits people for short-term events, casting a wide net over people who may be interested in volunteering for a "good cause", but may not be particularly drawn towards mapping in OSM. This means that they will put in work in the short-term, but likely not stick around without a structured drive. This notion is supported by looking at the average labour hours within the first 48 hours of the different drives; even with the lowest retention rates, Missing Maps has more short-term labour hours than YouthMappers, and even more so than the general OSM cohort (more than double the latter). These numbers also mirror the ones in Dittus et al., where Missing Maps had the most labour hours out of all the cohorts studied (along with the lowest short-term retention, as already mentioned). Mappers in the general OSM cohort are more likely to have joined out of a personal interest rather than recruited by a drive. Our results show that although they may not have the incentive or pressure to contribute a lot within the first few days of joining, they are more likely to return, and contribute over a longer period of time.

These conclusions hold when comparing Missing Maps to the general cohort, but the question remains of why YouthMappers, another mapping project that holds open mapping drives, lies between the two other projects, rather than having similarly low retention rates to Missing Maps. The reason may lie in the incentive structure for YouthMappers compared to the other two. While a large portion of its recruits may come from short-term drives (like Missing Maps), YouthMappers is primarily based around university clubs, so it has a solid underlying pool of contributors to draw from in the long-term [12]. Students interested in joining these clubs are more likely similar to the general cohort in having an above-average interest in mapping and OSM. The inherent structure of being in a club with recurrent meetings, and also the social value in working with the same group of people over time can also lead to higher retention rates. But since YouthMappers also hosts open drives similar to Missing Maps as mentioned, many recruits are also the sort of short-term recruits that contribute to Missing Maps drives, so the pool of long-term university contributors may be diluted by short-term volunteers.

This offers a potentially interesting model of collaboration. Rather than having purely long-term or short-term contributors, Ransbotham and Kane [10] demonstrated that in Wikipedia's case, an article having a mix of both new and old contributors increased its chance of becoming a featured article while decreasing its chance of being demoted. This model may merit further exploration in humanitarian contexts, where a solid, relatively long-term team can form the backbone of an operation to ensure continuity in vision and efforts, while also being able to host open drives and draw from a much larger pool of people. While the general cohort has the best retention rate overall, it's primarily composed of individuals contributing to whichever effort interests them at the time, so it's not necessarily a comparable model for ensuring focused effort on a particular project.

YouthMappers also has the advantage of its members spending a lot of time learning about the YouthMappers mission and its values, and thus feeling a strong sense of affinity towards this organization—in contrast to recruits from other organizations. The benefits of such engagement is supported by the conclusions from Cox et al. [4]: identifying with the values of an organization is strongly associated with better engagement and retention. Their main recommendation for increasing engagement and retention of volunteers to online projects is to spend time teaching new recruits about the core values of the organization. Similarly, other research within online collaborative communities such as Wikipedia [8] has demonstrated that incorporating social support mechanisms that pave the way for newcomer inclusion into the community, through mentorship opportunities and engagement with experienced users, can lead to higher rates of new user retention. This may be accomplished more easily over time, such as through scheduled meetings throughout the year, rather than through blitz mapathons.

These suggestions should not be taken as the only way to increase long-term engagement; rather they can be seen as additional tools that can be added to a project organizer's repertoire in order to increase long-term retention. Although these guidelines can be useful for new projects, or established projects that are willing to make perhaps difficult structural changes, it may not be practical for every sort of project. In this case, other guidelines from prior works can be used, such as focusing on ways to engage new users within their first 6 months of joining, as Bégin et al.'s conclusions point to that being a crucial period [3], or focusing more on relating the core values of a project to recruits, as Cox et al. suggests [4]. It can also be a good idea to take some time to focus recruitment efforts on more likely candidates; as Karumur [7] finds, recruiting from people who have prior volunteer experience, especially if they have worked on your project without necessarily having been a part of it yet, is positively correlated with retention. This last point may still be easier for organizations with YouthMapper's structure though; as many of their mapathon recruits may be students or people local to campus, it could be easier to identify promising potentials.

Although our results for Missing Maps differed from the original paper, many of the overall conclusions we draw are similar. Dittus et al. hypothesized that Missing Maps had the best long-term retention out of their cohorts because they had the best long-term engagement plan with mapathons. As we have noted above, YouthMappers, which wasn't considered in the initial paper has

well developed community engagement strategies. Another aspect to consider when comparing retention rates from pre- to post-2016 is the overall rise in new contributors. Although the original Dittus et al. paper was published in 2016, many of the drives considered were from 2014, where the number of new contributors for each drive was the lowest. The growth of new contributors has almost doubled in the years since, and retention rates overall have dipped from the pre-2016 numbers. This may be explained by the fact that most of the people from the original study were "early-adopters" as compared to the current users. Early-adopters may be more likely to be interested in GIS and these types of humanitarian efforts in general, so as to be in spaces where they were more likely to hear about OSM in the first place. As OSM has matured, it has been able to attract many more volunteers, but many of these new volunteers may not be as committed or as interested as early-adopters, and so their retention rate is not as high. These factors could be subject to future research, perhaps by surveying users for their reasons for contributing to OSM in the current era, as a follow-up to Budhathoki et al. [2].

6 CONCLUSION

In this paper, we revisited a prior study of contributor retention in OpenStreetMap using additional cohorts and an updated time period. Doing so helped to add further context to the important findings of the original paper and deepen our understanding of some of the dynamics at play. Our data suggests that events like mapathons that have been used so successfully by the Missing Maps project to create new data and introduce new people to the platform may not be well suited for attracting and retaining long-term members of the community. This contrasts with YouthMappers, whose long-term support system is built up of the many local university chapters who encourage sustained engagement. Recruitment in the OpenStreetMap community has increased significantly since the original paper was published, but as contributor retention overall remains quite low, further attention to these research questions and efforts to develop better systems for newcomer retention are warranted.

REFERENCES

- [1] Jennings Anderson, Robert Soden, Kenneth M Anderson, Marina Kogan, and Leysia Palen. 2016. EPIC-OSM: A software framework for OpenStreetMap data analytics. In *2016 49th Hawaii International Conference on System Sciences (HICSS)*. IEEE, 5468–5477.
- [2] Nama R. Budhathoki and Caroline Haythornthwaite. 2013. Motivation for Open Collaboration: Crowd and Community Models and the Case of OpenStreetMap. *American Behavioral Scientist* 57, 5 (2013), 548–575. <https://doi.org/10.1177/0002764212469364> arXiv:<https://doi.org/10.1177/0002764212469364>
- [3] Daniel Bégin, Rodolphe Devillers, and Stéphane Roche. 2018. The life cycle of contributors in collaborative online communities -the case of OpenStreetMap. *International Journal of Geographical Information Science* 32, 8 (2018), 1611–1630. <https://doi.org/10.1080/13658816.2018.1458312> arXiv:<https://doi.org/10.1080/13658816.2018.1458312>
- [4] Joe Cox, Eun Young Oh, Brooke Simmons, Gary Graham, Anita Greenhill, Chris Lintott, Karen Masters, and Jamie Woodcock. 2018. Doing Good Online: The Changing Relationships Between Motivations, Activity, and Retention Among Online Volunteers. *Nonprofit and Voluntary Sector Quarterly* 47, 5 (2018), 1031–1056. <https://doi.org/10.1177/0899764018783066> arXiv:<https://doi.org/10.1177/0899764018783066>
- [5] Martin Dittus, Giovanni Quattrone, and Licia Capra. 2016. Analysing Volunteer Engagement in Humanitarian Mapping: Building Contributor Communities at Large Scale. In *Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing* (San Francisco, California, USA) (CSCW '16). Association for Computing Machinery, New York, NY, USA, 108–118. <https://doi.org/10.1145/2818048.2819939>
- [6] R.Stuart Geiger and Aaron Halfaker. 2013. Using edit sessions to measure participation in wikipedia. 861–870. <https://doi.org/10.1145/2441776.2441873>
- [7] Raghav Pavan Karumur, Bowen Yu, Haiyi Zhu, and Joseph A. Konstan. 2018. *Content is King, Leadership Lags: Effects of Prior Experience on Newcomer Retention and Productivity in Online Production Groups*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3173574.3174080>
- [8] Jonathan T. Morgan and Aaron Halfaker. 2018. Evaluating the Impact of the Wikipedia Teahouse on Newcomer Socialization and Retention. In *Proceedings of the 14th International Symposium on Open Collaboration* (Paris, France) (*OpenSym '18*). Association for Computing Machinery, New York, NY, USA, Article 20, 7 pages. <https://doi.org/10.1145/3233391.3233544>
- [9] Leysia Palen, Robert Soden, T Jennings Anderson, and Mario Barrenechea. 2015. Success & scale in a data-producing organization: The socio-technical evolution of OpenStreetMap in response to humanitarian events. In *Proceedings of the 33rd annual ACM conference on human factors in computing systems*. 4113–4122.
- [10] Sam Ransbotham and Gerald C Kane. 2011. Membership turnover and collaboration success in online communities: Explaining rises and falls from grace in Wikipedia. *Mis Quarterly* (2011), 613–627.
- [11] Aaron Shaw and Eszter Hargittai. 2018. The Pipeline of Online Participation Inequalities: The Case of Wikipedia Editing. *Journal of Communication* 68, 1 (02 2018), 143–168. <https://doi.org/10.1093/joc/jqx003> arXiv:<https://academic.oup.com/joc/article-pdf/68/1/143/24216213/jqx003.pdf>
- [12] Patricia Solis, Sushil Rajagopalan, Lily Villa, Maliha Binte Mohiuddin, Ebenezer Boateng, Stellamaris Wavamunno Nakacwa, and María Fernanda Peña Valencia. 2020. Digital humanitarians for the Sustainable Development Goals: YouthMappers as a hybrid movement. *Journal of Geography in Higher Education* 0, 0 (2020), 1–21. <https://doi.org/10.1080/03098265.2020.1849067> arXiv:<https://doi.org/10.1080/03098265.2020.1849067>
- [13] YouthMappers. 2021. *About Us*. <https://www.youthmappers.org/about-us>